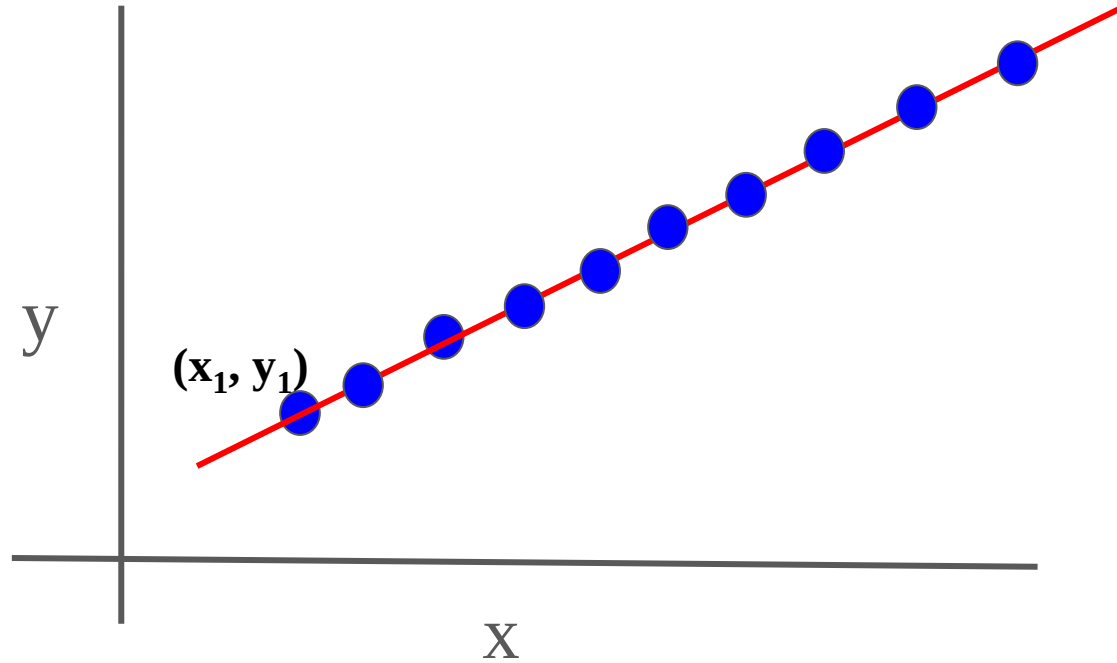


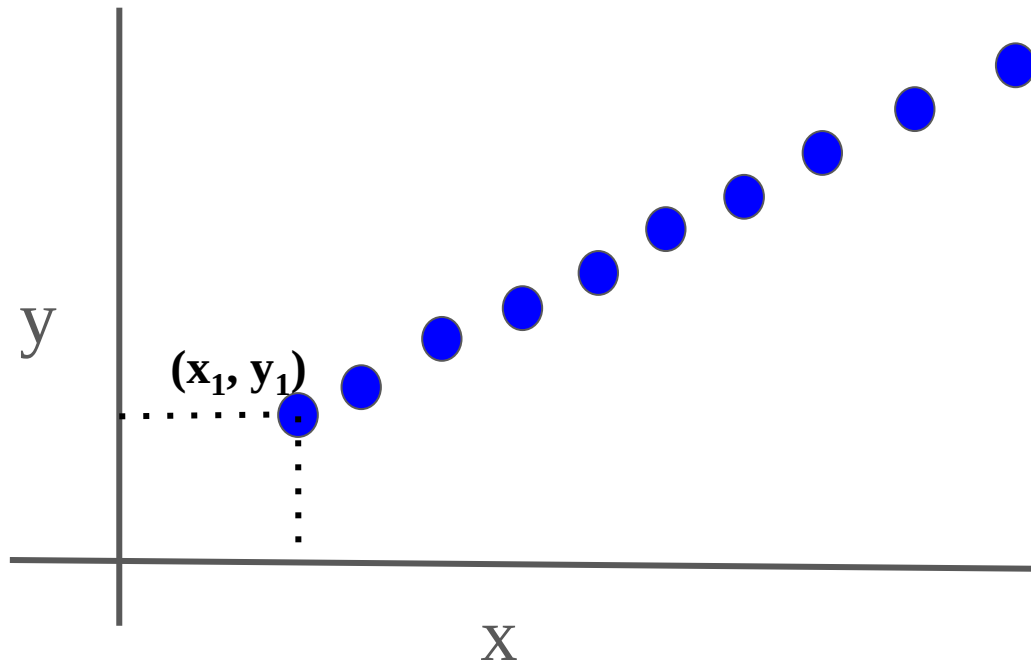
Linear Regression 1

Prof. Arun Chauhan, CSE
GEU, Dehradun

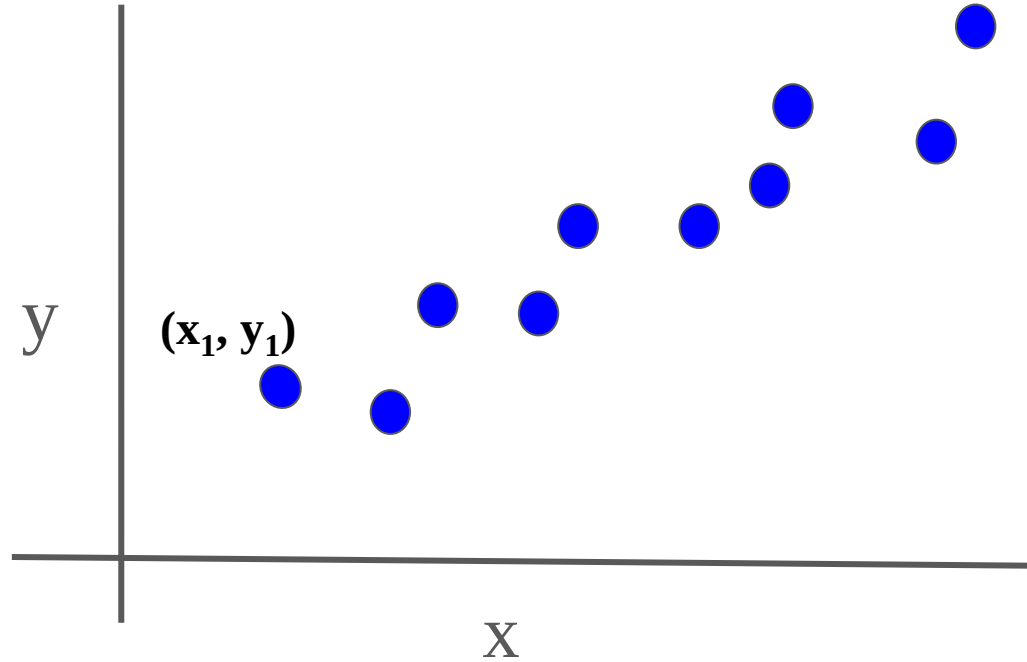
Curve fitting



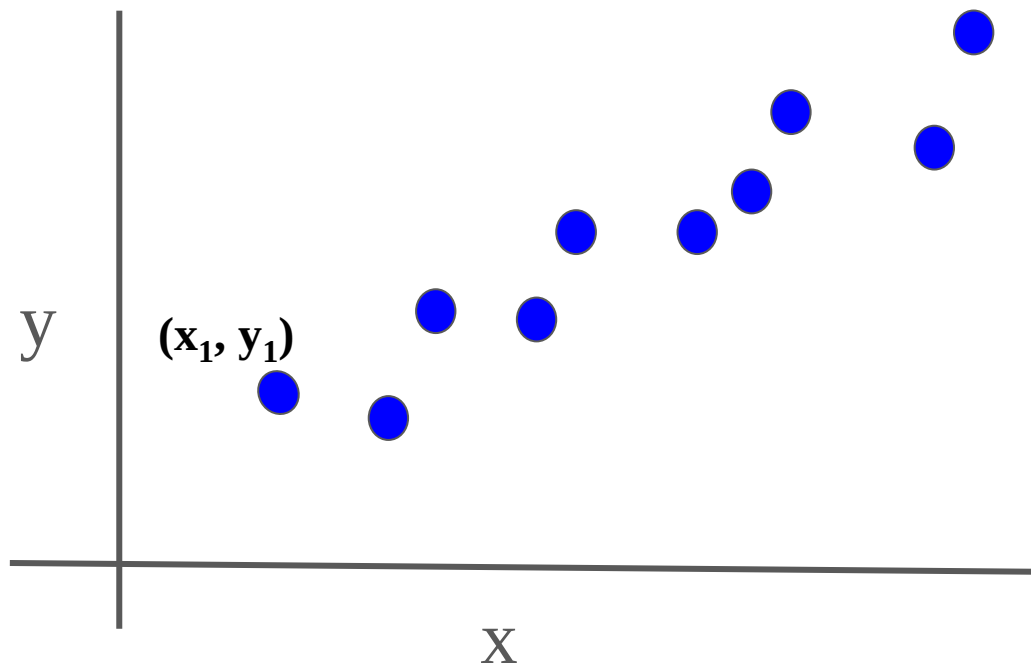
When function is not known !



Presence of noise



Can we recover the true function?



Let's try

First, we need to choose the function in general

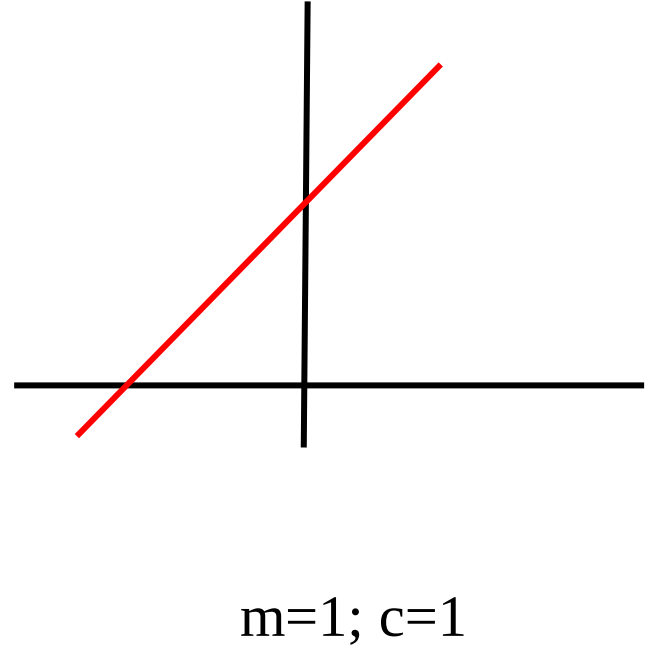
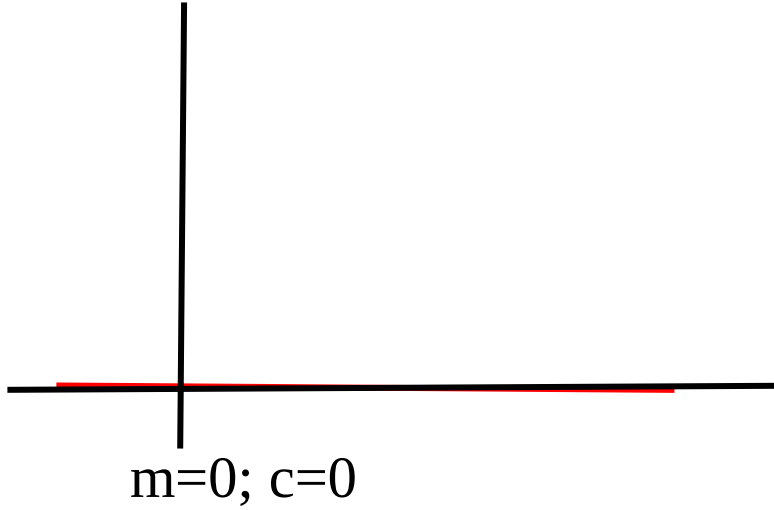
What does that mean?

Equation of line

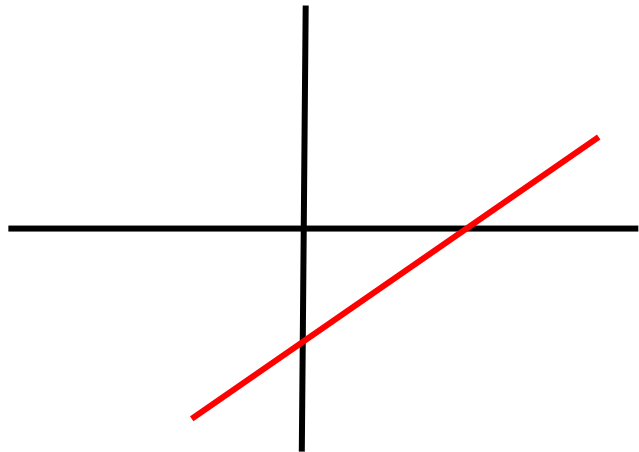
$$y = m * x + c$$

*For different values of **m** and **c**, we have
different lines*

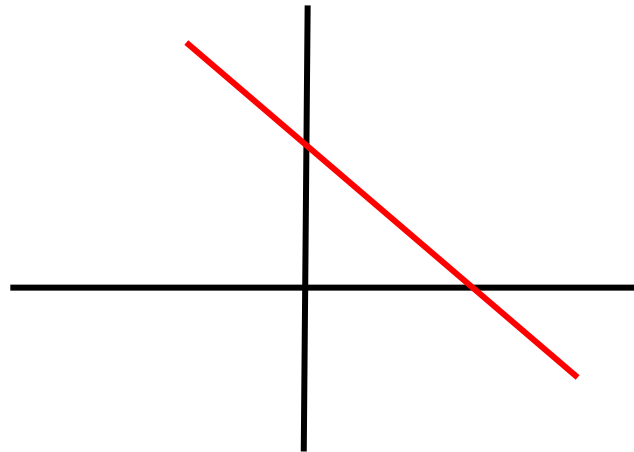
For example:



Example cont.



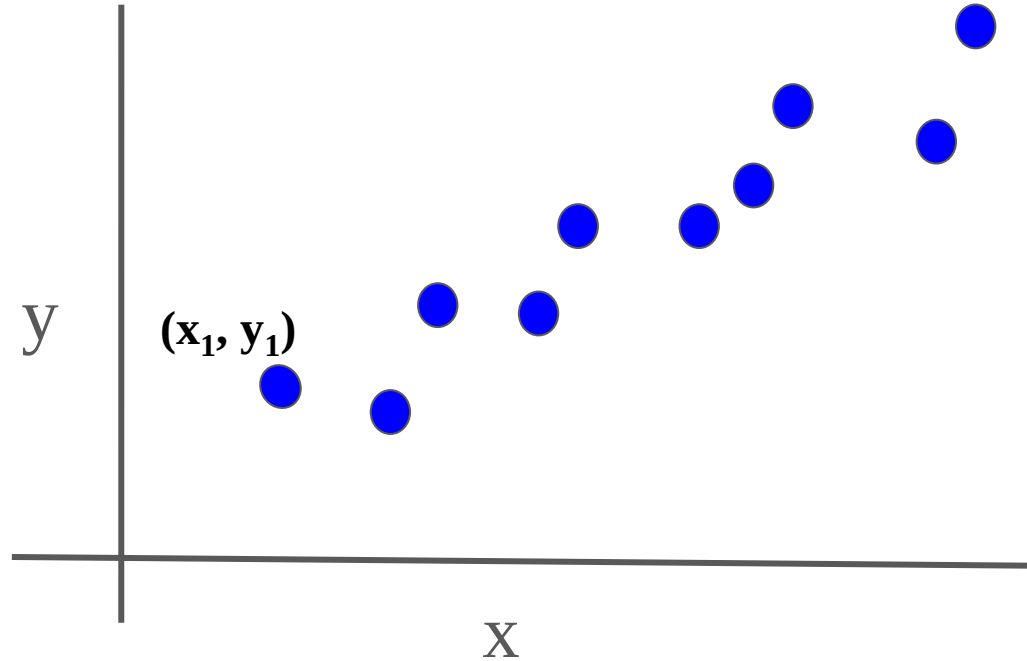
$$m=1; c=-1$$



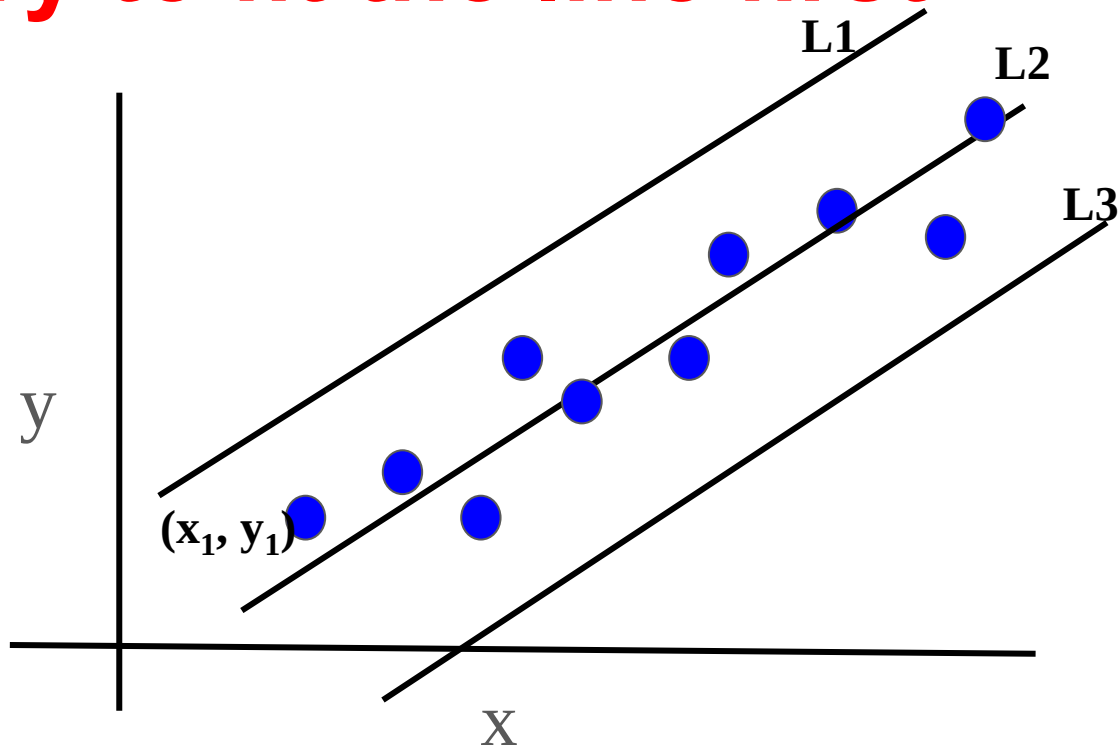
$$m=-1; c=1$$

*Let's come back to the
problem*

Can we recover the true function?



Let's try to fit the line first



Which line is best L1, L2, and L3?

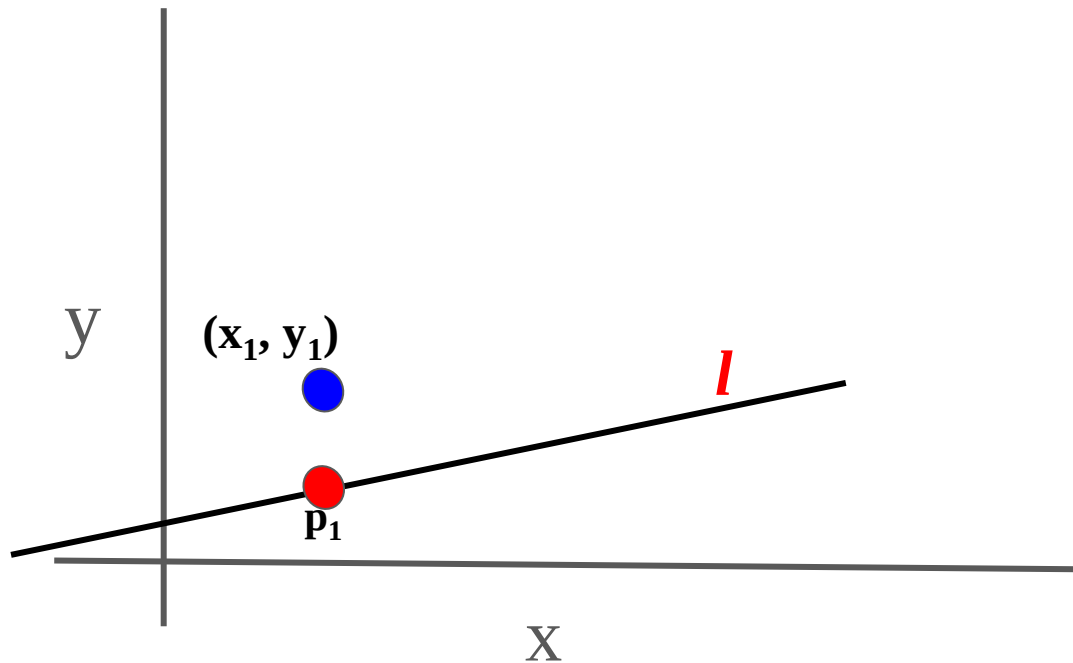
How to do it in computer?

We need to choose the optimal value of m & c

For any given line l defined by m and c

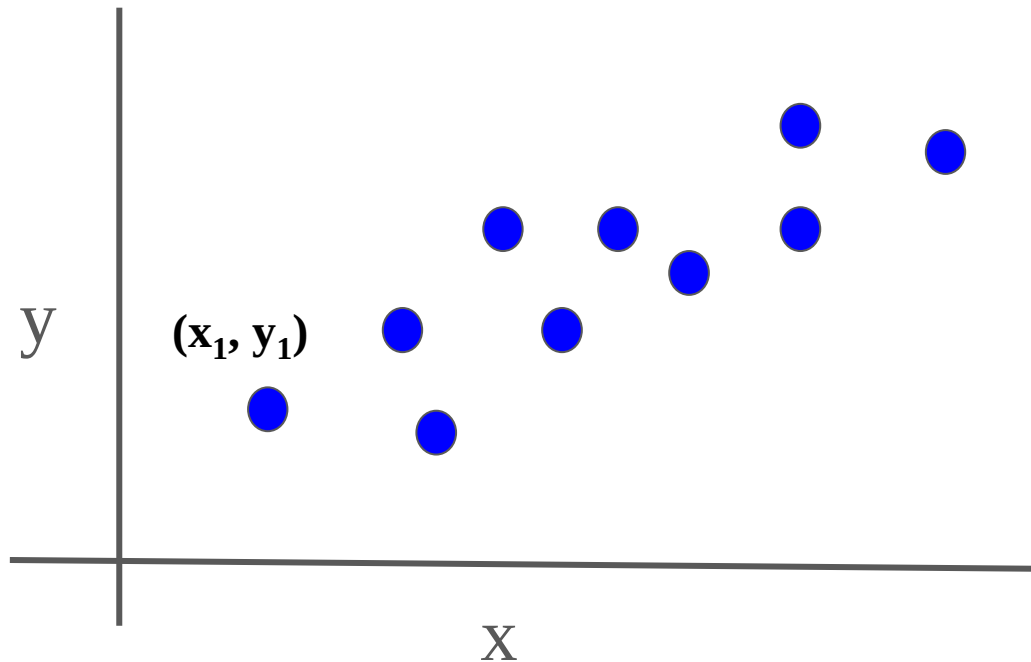
$$m * x + c = p$$

p is the prediction.



*Our purpose is to make the value of **p** as close to the value of **y** for better prediction*

Can we recover the true function?



We have 10 points:

Actual	Target	Error
$mx_1 + c = p_1$	y_1	$(y_1 - p_1)$
$mx_2 + c = p_2$	y_2	$(y_2 - p_2)$
.	.	.
.	.	.
.	.	.
$mx_{10} + c = p_{10}$	y_{10}	$(y_{10} - p_{10})$
	.	

Now, we take the square of the error.

Why square of the error?

Because +ve and -ve errors can cancel out each other.

Therefore;

Actual	Target	Error
$mx_1 + c = p_1$	y_1	$(y_1 - p_1)$
$mx_2 + c = p_2$	y_2	$(y_2 - p_2)$
.	.	.
.	.	.
.	.	.
$mx_{10} + c = p_{10}$	y_{10}	$(y_{10} - p_{10})$

$$\text{Total error} = (y_1 - p_1)^2 + (y_2 - p_2)^2 + \dots + (y_{10} - p_{10})^2$$

Alternatively

$$\text{Total error} = (\mathbf{y}_1 - \mathbf{p}_1)^2 + (\mathbf{y}_2 - \mathbf{p}_2)^2 + \cdots \cdots \cdots + (\mathbf{y}_{10} - \mathbf{p}_{10})^2$$

$$\text{Total error} = \sum_{\{i=1\}}^n (\mathbf{y}_i - \mathbf{p}_i)^2$$

Our **objective is to minimize the error !!!**

Simple trick from Calculus

Minimization of function:

$$\nabla f = 0$$

Take the derivative of the function and put it equal to zero

Hence, we get the optimal value of m & c

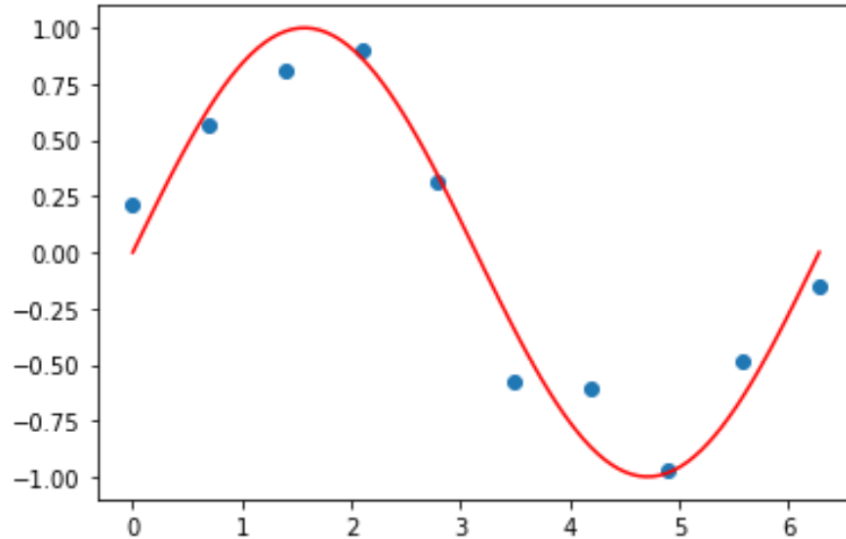
$$y = \mathbf{m} * x + \mathbf{c}$$

**We can use this (m,c) for predicting the output
for the new values of x**

CODE

[https://github.com/aruntakhur/GEU_ML_2024/blob/
main/LinearRegressionLab1GEU2024.ipynb](https://github.com/aruntakhur/GEU_ML_2024/blob/main/LinearRegressionLab1GEU2024.ipynb)

In next class, we will try to get a nonlinear function for scattered data points.



Thanks